

MIKKILI KOUSHIK

Python Developer

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PROFESSIONAL SUMMARY

Final-year B.Tech Computer Science (AI/ML) student with two internships and hands-on experience building real-time computer vision and NLP applications in Python, including FastAPI-integrated and scikit-learn-based projects. Strong foundation in Data Structures & Algorithms, PyTorch, scikit-learn, OpenCV, and MediaPipe, with a track record of shipping working prototypes independently and within agile teams.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, HTML, CSS

Frameworks & Tools: PyTorch, Scikit-learn, FastAPI, Flask, React.js, NLTK, Tailwind CSS

AI/ML & CV: OpenCV, MediaPipe, Hugging Face Transformers, Computer Vision, NLP, Gesture Recognition, Machine Learning

Database: MySQL

Concepts: OOP, Data Structures & Algorithms, Machine Learning, REST API Development

OS: Linux, Windows

PROJECTS

Air Mouse — Real-Time Hand Gesture Controller ([GitHub](#))

Python · OpenCV · MediaPipe · pyautogui · NumPy

- Built a real-time contactless mouse using MediaPipe hand tracking (21 landmarks) at 30 FPS, solving jitter with NumPy-based smoothing.
- Implemented pinch-to-click, scrolling, and gesture-driven volume control with cooldown logic and a live on-screen overlay.

Speech Translation & Emotion Detector ([GitHub](#))

Python · SpeechRecognition · Vosk · langdetect · Hugging Face Transformers · BART

- Built a multilingual voice assistant combining Google Speech Recognition (online) and Vosk (offline), with automatic language detection and translation to English.
- Implemented keyword-based emotion detection and used the BART transformer to summarize long speech inputs.

AI Virtual Assistant for Desktop ([GitHub](#))

Python · FastAPI · Vosk · pyttsx3 · MediaPipe · OpenCV · pandas · matplotlib · tkinter

- Developed a dual-mode voice assistant that auto-switches between online and offline speech recognition, with the Air Mouse module integrated for gesture-based control.
- Added voice-driven CSV analysis with spoken summaries and automatic chart generation using matplotlib.
- Exposed core assistant functions as REST endpoints using FastAPI, enabling external triggering of voice commands and CSV analysis.

Sentiment Analysis Tool ([GitHub](#))

Python · NLTK · Scikit-learn · NLP — Internship Project @ Slash Mark IT Solutions

- Built an NLP pipeline to classify customer reviews as positive or negative using NLTK preprocessing (tokenization, stopword removal, feature extraction).
- Trained and evaluated classification models using scikit-learn, comparing accuracy, precision, and recall to select the best-performing approach.

INTERNSHIP EXPERIENCE

AI/ML Intern — Micro Information Technology Services

Mullana, Ambala | May – Jun 2025 | Certificate ID: MITS136352

- Built an offline multilingual speech recognition system using Vosk, removing dependency on internet connectivity.
- Implemented automatic language detection achieving ~75% accuracy across Hindi, Telugu, and English samples.
- Developed a speech translation pipeline converting detected languages to English for downstream processing.

Software Development Intern — Slash Mark IT Solutions (OPC) Pvt. Ltd.

Oct – Dec 2024

- Built a multilingual speech processing system combining Google Speech Recognition and Vosk for online/offline transcription.
- Implemented language detection and translation using langdetect, plus keyword-based emotion analysis.
- Integrated Hugging Face BART (facebook/bart-large-cnn) to summarize long speech inputs across multiple languages.

EDUCATION

B.Tech – Computer Science Engineering (AI/ML) 2022 – 2026

Dhanalakshmi Srinivasan University | CGPA: 7.59

CERTIFICATIONS

AI/ML Internship Certificate (Micro Information Technology Services) || Natural Language Processing (Infosys) || Python Programming (Scaler)